

System for Reaching Prime Numbers Trough Lines

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An insight looking trough prime numbers graphical arrangement

1 I

is possible to explain visually why some geometrical patterns appear in certain prime number representations. Looking trough geometry it can be possible to explain the distribution of complex patterns. The course of ideas will follow until drawing over the known polar-spiral structure, lines, being the result of projecting residual classes.

1.1 T

the prime numbers are commonly represented with the equations of the polar-spiral graph, where every prime number with radio $*p^*$ and angle $*p^*$:

$$z_p = pe^{ip} \quad (1)$$

And this behaviour come from a polar representation of integers, with a gap made between certain integers and the distribution of prime numbers in residual clases coprimes of module (N)

1.2 E

every natural number can be colocated in the plane trough $[z_n = ne^{in}]$ This equals assaying $[x_n = n \cos(n)][y_n = n \sin(n)]$ Where the number (n) determinessimultaneously the distancetocenter and the rotation angle. $[z_p = pe^{ip}]$ Where (p) passes by the prime numbers.

2 Partitions on the spiral

For concreting the multiple-partitions spiral structure, we choose a module (N), reaching that every natural number can be calssified according to its residue. $[n \equiv r \pmod{N}]$ And it's found that numbers of one residue $r+kN$ Putting then in a polar-spiral $[z_{r+kN} = (r+kN)e^{i(r+kN)}]$ And separating the angular terms : $[z_{r+kN} = e^{ir}(r+kN)e^{ikN}]$ Stabishing the relations with the near by multiples, we have that if (N) is near a multiple of (2) $[N \approx 2\pi m]$ and by that : $[e^{ikN} \approx 1]$ Implies that the numbers of onesame residual classtendstobenear the same angular

2.1 Introduction of Dirichet Theorem

The tendency of the residual classes for obtaining infinite prime numbers don't apply to all of them. If a residue (r) shares a factor with (N), so a lot of numbers of the form $[r+kN]$ Will be compound. For that are considered specially the classes where $[\gcd(r, N) = 1]$ Being them the residual coprimal classes with (N).

$[r, r + N, r + 2N, r + 3N, \dots]$ The lines correspond to residual classes $(r \pmod{N})$, with $\gcd(r, N) = 1$

2.2 Module 44

Remembering the example given by the initial explanation where is inspired this method, we have that the number 44 is near of seven whole spins, $[44 \approx 7(2\pi)]$ For that the residual classes of module(44) generates modulo 44 as : $[\varphi(44) = 20]$ There are twenty permitted classes that correspond to 20 lines expected in the visualization.

2.3 Error Tracking

As the alignment ain't perfect because (N) doesn't equal exactly a multiple of (2π) . It can be written $[N = 2\pi m + \varepsilon]$ Where (ε) is the angular error. Every time we step ahead one unit in the same residual class, the error accumulates $[k\varepsilon]$ If (ε) is little, the line seems straight during very long time. If it is larger, the line bends faster. Soon, the straightness $[N \approx 2\pi m]$

/section Proposed method Formulating the analysis method, we find that, in first, we graph the prime numbers through the function: $[z_p = pe^{ip}]$ Then, we choose a modulo (N) near a multiple of (2π) : $[N \approx 2\pi m]$ And we divide the numbers in residual classes : $[p \equiv r \pmod{N}]$ We identify the coprimal classes : $[\gcd(r, N) = 1]$ And trace geometrical guides to that classes, compare the visual distribution of the primes with the permitted $[\varepsilon = N - 2\pi m]$ We analyze what modules produce more persistent lines.